

# WEEK 12 - PHP MVC FRAMEWORK

## LAB ACTIVITY – CodeIgniter

### Syllabus:

- Implementing JQuery AJAX
- Adding navigation menu
- Composer
- PHPMailer

### Implementing JQuery AJAX

Mengimplementasikan teknik JQuery AJAX pada CodeIgniter dimungkinkan. Pada *project* yang sudah kita miliki, pengimplementasian akan kita lakukan pada tombol Edit dan Delete.

1. Untuk dapat menggunakan *library JQuery* maka pastikan kita sudah menyisipkannya pada *file view header\_view.php* dan menyediakan *JQuery resource file* di folder assets.

```
<!DOCTYPE html>
<html>
  <head>

    <link rel="stylesheet" href="<?php echo base_url('assets/bootstrap-4.0.0-
dist/css/bootstrap.min.css'); ?>">

    <script src="<?php echo base_url('assets/jquery-3.2.1.min.js'); ?>"></script>
    <script src="<?php echo base_url('assets/bootstrap-4.0.0-
dist/js/bootstrap.min.js'); ?>"></script>

    <title><?=$title;?></title>

  </head>

  <body>
```

2. Selanjutnya lakukan *testing* untuk memastikan apakah *library JQuery* sudah berfungsi dengan baik. Untuknya pada *file view mahasiswa\_view.php* tambahkan *script* berikut:

```
<script>

$(this).ready(function() {

    alert('Holaaaaa...');

})

</script>
```



Data Mahasiswa

Tambah

| NIM       | Nama               | Email          | Action                            |
|-----------|--------------------|----------------|-----------------------------------|
| 412017001 | Joshua Christian A | josh@gmail.com | <div>Edit</div> <div>Delete</div> |
| 412017002 | Kevin Nathanael    |                | <div>Edit</div> <div>Delete</div> |
| 412017003 | Gabriel Nathanael  |                | <div>Edit</div> <div>Delete</div> |

Holaaaa...  

OK

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Setelah berhasil, perintah *alert* tadi dapat dihilangkan atau ditandai sebagai komentar.

### Updating process

- Selanjutnya untuk proses edit siapkan **formEdit** dengan menggunakan *modal mdEdit*. Pada *file view mahasiswa\_view.php* tambahkan *script* sebagai berikut:

```
<div id="mdEdit" class="modal" tabindex="-1" role="dialog">
  <div class="modal-dialog" role="document">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title"></h5>
        <button type="button" class="close" data-dismiss="modal" aria-label="Close">
          <span aria-hidden="true">&times;</span>
        </button>
      </div>

      <form id="formEdit">
        <div class="modal-body">
          <div class="form-group">
            <label><strong>NIM</strong></label>
            <input type="text" class="form-control" id="nim" name="nim" value="">

            <label><strong>Nama</strong></label>
            <input type="text" class="form-control" id="nama" name="nama" value="">
          </div>
        </div>
      </form>
    </div>
  </div>
</div>
```



```

<label><strong>Alamat</strong></label>
<input type="text" class="form-control" id="alamat" name="alamat"
value="">

<label><strong>Email</strong></label>
<input type="text" class="form-control" id="email" name="email"
value="">

<label><strong>Kontak</strong></label>
<input type="text" class="form-control" id="kontak" name="kontak"
value="">
</div>
</div>
<div class="modal-footer">
<button type="submit" id="btnSave" class="btn btn-primary">Save
changes</button>
<button type="button" id="btnCancel" class="btn btn-secondary" data-
dismiss="modal">Cancel</button>
</div>
</form>

</div>
</div>
</div>

```

4. Selanjutnya *modal* mdEdit akan kita munculkan setiap kali tombol Edit ditekan. Untuknya lakukan modifikasi pada *syntax* yang ada pada file view **mahasiswa\_view.php** sebagai berikut:

```

<div class="container-fluid">

<div class="row">

<div class="col-12">

<h2 class="text-center mt-3"><?=$title; ?></h2>
<a href="<?php echo base_url('mahasiswa/tambah'); ?>" type="button"
class="btn btn-primary mb-3">Tambah</a>

<table id="myTable" class="table">
<thead>
<td>NIM</td>
<td>Nama</td>
<td>Email</td>
<td>Action</td>
</thead>

<?php

foreach($mahasiswa as $row){
echo '<tr>';
echo '<td>'.$row->nim.'</td>';

```



```

        echo '<td>'.$row->nama.'</td>';
        echo '<td>'.$row->email.'</td>';
        echo '<td><button type="submit" data-id="'.$row->nim.'" id="btnEdit"
data-toggle="modal" data-target="#mdEdit" class="btn btn-primary">Edit</button>
<button class="btn btn btn-danger">Delete</button></td>';

        echo '</tr>';
    }
    ?>

</table>

</div>

</div>

</div>

```

Lakukan *testing* apakah *modal* berhasil dimunculkan pada saat tombol Edit diklik?

| NIM       | Nama               | Action                                        |
|-----------|--------------------|-----------------------------------------------|
| 412017001 | Joshua Christian A | <button>Edit</button> <button>Delete</button> |
| 412017002 | Kevin Nathanael    | <button>Edit</button> <button>Delete</button> |
| 412017003 | Gabriel Natanael   | <button>Edit</button> <button>Delete</button> |

**Edit**

NIM

Nama

Alamat

Email

Kontak

Save changes Close

- Sekarang kita perlu menangani *onclick event* dari **btnEdit** agar *data record* terkait dapat ditampilkan di *modal*. Tambahkan perintah *javascript* berikut pada file view **mahasiswa\_view.php**:

```

$(this).ready(function() {

    $(this).on('click', '#btnEdit', function() {

```

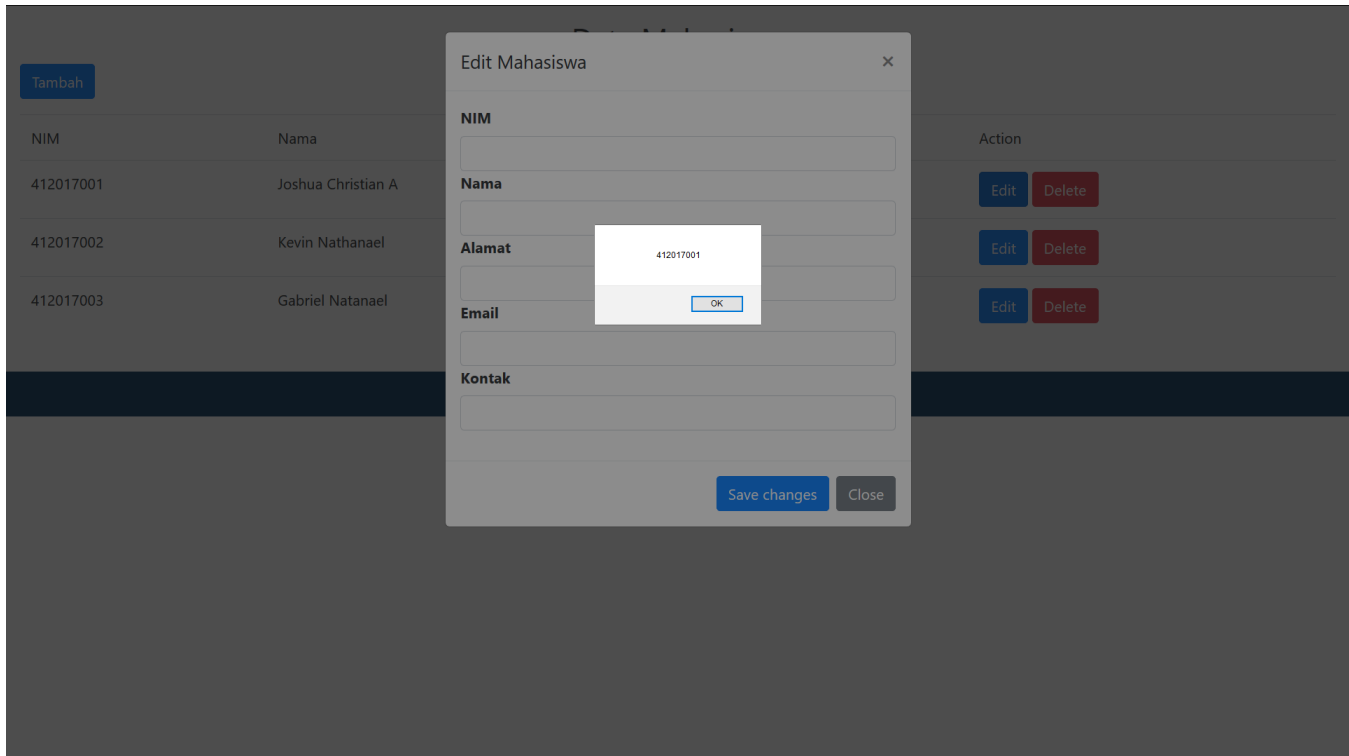


```
$('.modal-title').html('Edit Mahasiswa');

var nim = $(this).data('id');
alert(nim);

})
```

Perhatikan bahwa kita menyimpan nilai nim dari setiap *record* yang ditampilkan pada tabel dalam atribut **data-id**. Di sini yang kita lakukan adalah mengambil nilai tersebut lalu menampilkannya dalam bentuk pesan ke layar.



- Sampai di sini kita perlu menyiapkan sebuah *method* di *class Controller* untuk menghandel parameter nim yang nanti akan kita kirim. Untuknya pada *class Controller Mahasiswa* tambahkan satu *method* baru sebagai berikut:

```
public function getMhs() {

    $nim=$this->input->post('nim');

    $data = $this->mahasiswa->get_mahasiswa_by_nim($nim);

    echo json_encode($data);

}
```



Perhatikan bahwa parameter nim selanjutnya akan dikirimkan ke method **get\_mahasiswa\_by\_nim()** yang sebelumnya sudah pernah kita buat di *class Model Mahasiswa\_model.php*. Data kembalian selanjutnya ditampung dalam variable `$data` dan dicetak dalam format *json*.

7. Selanjutnya implementasikan *Jquery AJAX* untuk menghadirkan data pada modal dengan menambahkan perintah *javascript* berikut pada *file view mahasiswa\_view.php*:

```
$(this).ready(function() {

    $(this).on('click', '#btnEdit', function() {

        $('#modal-title').html('Edit Mahasiswa');
        var nim = $(this).data('id');

        $.ajax({
            type: 'POST',
            url: '<?php echo base_url("mahasiswa/getMhs"); ?>',
            data: {nim: nim},
            dataType: 'json'
        }).done(function(data) {

            $('#id').val(data.id);
            $('#nim').val(data.nim);
            $('#nama').val(data.nama);
            $('#alamat').val(data.alamat);
            $('#email').val(data.email);
            $('#kontak').val(data.kontak);

        });

    });

});
```



Tambah

| NIM       | Nama               |
|-----------|--------------------|
| 412017001 | Joshua Christian A |
| 412017002 | Kevin Nathanael    |
| 412017003 | Gabriel Natanael   |

×

Edit Mahasiswa

NIM

Nama

Alamat

Email

Kontak

Save changes

Close

Action

Edit

Delete

Edit

Delete

Edit

Delete

8. Berikutnya untuk menangani proses *update data* maka tambahkan perintah *javascript* berikut pada *file view mahasiswa\_view.php*:

```
$(this).ready(function() {

    $("#formEdit").submit(function(event) {

        event.preventDefault();

        $('.modal-footer').hide();

        var formData = $('#formEdit').serializeArray();

        console.log(formData);

    });

    $(this).on('click', '#btnEdit', function(){

        ...

    });

})
```



## Data Mahasiswa

Tambah

| NIM       | Nama               | Email          | Action                                                                                                                                                                                                              |
|-----------|--------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 412017001 | Joshua Christian A | jos@gmail.com  | <span style="background-color: #007bff; color: white; padding: 2px 5px; border-radius: 3px;">Edit</span> <span style="background-color: #dc3545; color: white; padding: 2px 5px; border-radius: 3px;">Delete</span> |
| 412017002 | Kevin Nathanael    | kev@gmail.com  | <span style="background-color: #007bff; color: white; padding: 2px 5px; border-radius: 3px;">Edit</span> <span style="background-color: #dc3545; color: white; padding: 2px 5px; border-radius: 3px;">Delete</span> |
| 412017003 | Gabriel Natanael   | abcd@xyz.ac.id | <span style="background-color: #007bff; color: white; padding: 2px 5px; border-radius: 3px;">Edit</span> <span style="background-color: #dc3545; color: white; padding: 2px 5px; border-radius: 3px;">Delete</span> |

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```

[3] log
> 0: object { name: "nim", value: "412017001" }
> 1: object { name: "nama", value: "Joshua Christian A" }
> 2: object { name: "alamat", value: "Jl. Jati Sore" }
> 3: object { name: "email", value: "jos@gmail.com" }
> 4: object { name: "kontak", value: "0876543210" }
length: 5
    
```

Pada gambar di atas kita menggunakan perintah **console.log()** untuk mencetak var formData ke *console browser* (bisa diaktifkan dengan tombol F12 lalu pilih *tab console*).

9. Berikutnya kita perlu menyiapkan *method* di *class Controller Mahasiswa* untuk menangani proses *update* ini:

```

public function updateMhs() {

    $data = array(
        'nim' => $this->input->post('nim'),
        'nama' => $this->input->post('nama'),
        'alamat' => $this->input->post('alamat'),
        'email' => $this->input->post('email'),
        'kontak' => $this->input->post('kontak'));

    $this->db->where('nim', $this->input->post('nim'));
    if($this->db->update('mahasiswa', $data))
        echo "Records Saved Successfully";
    else
        echo "Something wrong!";

}
    
```





10. Selanjutnya terapkan teknik *AJAX* untuk mengirimkan data ke *class Controller*:

```
$("#formEdit").submit(function(event) {  
  
    event.preventDefault();  
  
    $('.modal-footer').hide();  
  
    var formData = $('#formEdit').serializeArray();  
  
    console.log(formData);  
  
    $.ajax({  
        type: 'POST',  
        url: '<?php echo base_url("mahasiswa/updateMhs"); ?>',  
        data: formData,  
        async: false  
    }).done(function(data) {  
        console.log(data);  
  
        $('.modal-body').html(data);  
  
        $('.modal-footer').show();  
        $('.modal-footer #btnCancel').hide();  
        $('.modal-footer #btnSave').html('Close');  
        $(".modal-footer #btnSave").attr("id", "btnClose");  
  
        $(".modal-footer #btnClose").click(function() {  
            $('.close').click();  
            location.reload();  
        });  
  
    });  
  
});
```

Lakukan percobaan *update date* dan amati hasilnya.



Tambah

| NIM       | Nama               |
|-----------|--------------------|
| 412017001 | Joshua Christian A |
| 412017002 | Kevin Nathanael    |
| 412017003 | Gabriel Nathanael  |

✕

Edit Mahasiswa

NIM

412017003

Nama

Gabriel Omar Batistuta

Alamat

Jl. Sapoli Barat

Email

abcd@xyz.co.id

Kontak

0809000000

Save changes

Cancel

Action

Edit

Delete

Edit

Delete

Edit

Delete

Tambah

| NIM       | Nama               |               |
|-----------|--------------------|---------------|
| 412017001 | Joshua Christian A |               |
| 412017002 | Kevin Nathanael    | kev@gmail.com |
| 412017003 | Gabriel Nathanael  | abcd@xyz.com  |

✕

Edit Mahasiswa

Records Saved Successfully

Close

Action

Edit

Delete

Edit

Delete

Edit

Delete

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| Data Mahasiswa         |                        |                |                                             |
|------------------------|------------------------|----------------|---------------------------------------------|
| <a href="#">Tambah</a> |                        |                |                                             |
| NIM                    | Nama                   | Email          | Action                                      |
| 412017001              | Joshua Christian A     | jos@gmail.com  | <a href="#">Edit</a> <a href="#">Delete</a> |
| 412017002              | Kevin Nathanael        | kev@gmail.com  | <a href="#">Edit</a> <a href="#">Delete</a> |
| 412017003              | Gabriel Omar Batistuta | abcd@xyz.co.id | <a href="#">Edit</a> <a href="#">Delete</a> |

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## Deleting process

Dengan pendekatan yang sama kerjakan *deleting process* sebagai *lab activity* berikutnya.

## Adding Navigation Menu

11. Buat *file view* dengan nama **menu\_view.php** lalu isikan dengan *syntax* berikut:

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <a class="navbar-brand" href="#">UKRIDA</a>
  <button class="navbar-toggler" type="button" data-toggle="collapse" data-
target="#navbarTogglerDemo02" aria-controls="navbarTogglerDemo02" aria-
expanded="false" aria-label="Toggle navigation">
    <span class="navbar-toggler-icon"></span>
  </button>

  <div class="collapse navbar-collapse" id="navbarTogglerDemo02">
    <ul class="navbar-nav mr-auto mt-2 mt-lg-0">
      <li class="nav-item">
        <a class="nav-link" href="<?php echo base_url('mahasiswa'); ?>">Mahasiswa
      </li>
      <li class="nav-item">
        <a class="nav-link" href="">Program Studi</a>
      </li>
    </ul>

    </div>
  </nav>
```



12. Selanjutnya tambahkan pada setiap *method* dari *class Controller Mahasiswa* yang diperlukan. Misalkan pada *method index()*:

```
public function index(){
    $data['mahasiswa'] = $this->mahasiswa->get_mahasiswa()->result();
    $data['mahasiswa_array'] = $this->mahasiswa->get_mahasiswa_array();

    $header['title']= $this->mahasiswa->get_title();
    $this->load->view('header_view', $header);
    $this->load->view('menu_view', $data);
    $this->load->view('mahasiswa_view', $data);
    $this->load->view('footer_view');
}
```

| UKRIDA Mahasiswa Program Studi |                    |                |                                             |
|--------------------------------|--------------------|----------------|---------------------------------------------|
| Data Mahasiswa                 |                    |                |                                             |
| <a href="#">Tambah</a>         |                    |                |                                             |
| NIM                            | Nama               | Email          | Action                                      |
| 412017001                      | Joshua Christian A | josh@gmail.com | <a href="#">Edit</a> <a href="#">Delete</a> |
| 412017002                      | Kevin Nathanael    | kev@gmail.com  | <a href="#">Edit</a> <a href="#">Delete</a> |
| 412017003                      | Gabriel Nathanael  | abcd@xyz.com   | <a href="#">Edit</a> <a href="#">Delete</a> |

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## Composer

Selain menginstalasi CodeIgniter secara manual, kita juga dapat menggunakan composer. Composer merupakan Dependency Manager untuk PHP, sama halnya dengan NPM pada node.js.

Composer sangat dibutuhkan apabila kita bekerja dengan banyak *library*. Dalam hal ini composer akan membantu kita menginstal, men-download, meng-update, dan mencari *dependency* dari *library* yang digunakan.

Untuk dapat melakukan instalasi codeigniter 4 menggunakan composer, kita harus menginstal composer terlebih dahulu..

- Untuk Windows, Composer-Setup.exe dapat diunduh pada link: <https://getcomposer.org/download/>
- Untuk Linux/Unix/macOS, pada link: <https://getcomposer.org/doc/00-intro.md>

Untuk memastikan apakah composer telah terpasang di komputer kita, ketikkan perintah berikut pada *command prompt*:

```
Composer -v
```



```

C:\WINDOWS\system32\cmd.exe
C:\Users\Aurora\composer -e

Composer

Composer version 2.1.7 2021-06-09 16:31:20

Usage:
  command [options] [arguments]

Options:
  -h, --help                Display this help message
  -q, --quiet               Do not output any message
  -V, --version             Display this application version
  --ansi                    Force ANSI output
  --no-ansi                Disable ANSI output
  -n, --no-interaction      Do not ask any interactive question
  --profile                Display timing and memory usage information
  --no-plugins              Whether to disable plugins
  -d, --working-dir=WORKING-DIR
                           If specified, use the given directory as working directory.
  --cache                  Prevent use of the cache
  --verbose, --verbose=V   Increase the verbosity of messages: 1 for normal output, 2 for more verbose output and 3 for debug

Available commands:
  about                    Shows a short information about Composer.
  archive                 Creates an archive of this composer package.
  browse                  Opens the package's repository URL or homepage in your browser.
  clear-cache             Clears composer's internal package cache.
  check-platform-reqs    Check that platform requirements are satisfied.
  clear-cache             Clears composer's internal package cache.
  clear-autoload          Clears composer's internal package cache.
  config                 Sets config options.
  create-project          Creates new project from a package into given directory.
  depends                Shows which packages cause the given package to be installed.
  diagnose                Diagnoses the system to identify common errors.
  dump-autoload           Dumps the autoloader.
  dump-autoload-raw       Dumps the autoloader.
  exec                    Executes a vendored library/script.
  fund                    Discover how to help fund the maintenance of your dependencies.
  global                  Allow running commands in the global composer dir ($COMPOSER_HOME).
  help                   Displays help for a command
  home                   Opens the package's repository URL or homepage in your browser.
  i                       Installs the project dependencies from the composer.lock file if present, or falls back on the composer.json.
  info                   Shows information about packages.
  init                   Creates a basic composer.json file in current directory.
  install                Installs the project dependencies from the composer.lock file if present, or falls back on the composer.json.
  licenses               Shows information about licenses of dependencies.
  list                   Lists Commands
  outdated               Shows a list of installed packages that have updates available, including their latest version.
  packages               Shows which packages prevent the given package from being installed.
  reinstall              Uninstalls and reinstalls the given package names
  remove                Removes a package from the require or require-dev.
  require               Adds required packages to your composer.json and installs them.
  run                   Runs the scripts defined in composer.json.
  run-script             Runs the scripts defined in composer.json.
  search                 Searches for packages.
  self-update            Updates composer.phar to the latest version.
  self-update --no-interaction
                           Updates composer.phar to the latest version.
  show                  Shows information about packages.
  status                Shows a list of locally modified packages.
  suggestions            Shows package suggestions.
  update                Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  update --no-interaction
                           Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  update --no-progress
                           Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  update --quiet         Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  update --verbose       Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  update --no-progress
                           Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  update --quiet         Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  update --verbose       Upgrades your dependencies to the latest version according to composer.json, and updates the composer.lock file.
  
```

Selanjutnya kita akan menggunakan composer untuk mengirimkan email menggunakan Gmail.

1. Pada file **composer.json** tambahkan baris perintah berikut pada bagian require:  
**"phpmailer/phpmailer": "6.1"**
2. Lakukan update dengan mengetikkan perintah berikut pada command prompt:  
**composer update**

```

File Edit Selection View Go Run Terminal Help
• composer.json - ci-framework-4.1.1 - Visual Studio Code

EXPLORER
• composer.json
• ci-framework-4.1.1
  > app
  > public
  > system
  > vendor
  > writable
  > .env
  > .gitignore
  > composer.json
  > composer.lock
  > env
  > LICENSE
  > phpunit.xml.dist
  > README.md
  > spark

composer.json
{
  "name": "codeigniter4/framework",
  "type": "project",
  "description": "The CodeIgniter framework v4",
  "homepage": "https://codeigniter.com",
  "license": "MIT",
  "require": {
    "php": "^7.3|^8.0",
    "ext-curl": "*",
    "ext-intl": "*",
    "ext-json": "*",
    "ext-mbstring": "*",
    "kint-php/kint": "^3.3",
    "laminas/laminas-escaper": "^2.6",
    "psr/log": "^1.1",
    "phpmailer/phpmailer": "6.1"
  },
  "require-dev": {
    "codeigniter4/codeigniter4-standard": "^1.0",
    "fakerphp/faker": "^1.9",
    "mikey179/vfsstream": "^1.6",
    "phpunit/phpunit": "^9.1",
    "predis/predis": "^1.1",
    "squizlabs/php_codesniffer": "^3.3"
  }
}

TERMINAL
Generating autoload files
30 packages you are using are looking for funding.
Use the 'composer fund' command to find out more!
> CodeIgniter\ComposerScripts::postUpdate

C:\htdocs\ci-framework-4.1.1>

```



3. Selanjutnya buat file *Controller Mailer* yang berisi sebuah method **send()** untuk menghandle proses pengiriman email.

```
<?php namespace App\Controllers;

use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\SMTP;
use PHPMailer\PHPMailer\Exception;

class Mailer extends BaseController{

    public $to;
    public $subject;
    public $message;
    public $addCc;

    public $username='your_name';
    public $email='your_email';
    public $password='your_password';

    public function send(){

        $mail = new PHPMailer(true);

        try {
            $mail->isSMTP();
            $mail->Host      = 'smtp.googlemail.com';
            $mail->SMTPAuth  = true;
            $mail->Username  = $this->email;
            $mail->Password  = $this->password;
            $mail->SMTPSecure = 'ssl';
            $mail->Port      = 465;

            $mail->setFrom($this->email, '');

            $mail->addReplyTo($this->email, $this->username);
            $mail->isHTML(true);
            $mail->Subject = $this->subject;
            $mail->Body    = $this->message;

            if (!empty($this->to) && is_array($this->to)) {
                foreach ($this->to as $recipient) {
                    $mail->addAddress($recipient['email'], $recipient['name']);
                }
            } else $mail->addAddress($this->to);

            if (!empty($this->addCc) && is_array($this->addCc)) {
                foreach ($this->addCc as $recipient) {
                    $mail->addCc($recipient['email'], $recipient['name']);
                }
            } else $mail->addCc($this->addCc);

            $mail->send();
            return true;

        } catch (Exception $e) {
```



```
        $this->ErrorInfo=$e;  
        return false;  
    }  
  
}
```

4. Selanjutnya kita akan mengimplementasikan fitur ini pada controller **Registrasi > newAccount()** yang sudah ada. Sehingga perintah yang harus ditambahkan adalah sebagai berikut:

```
$sendEmail = new Mailer();  
  
$sendEmail->to="$this->request->getPost('username')";  
  
$sendEmail->subject='Registrasi';  
  
$sendEmail->message="  
    Dear ".$this->request->getPost('username').",<br><br>  
    You already registered on our system. Thanks.<br><br>Best regards";  
  
if($sendEmail->send()){  
    session()->setFlashData('success', 'Registration successfully.');} else  
    session()->setFlashdata('errors', ["We can\'t reach you by email!<br>Error:  
".$sendEmail->ErrorInfo]);  
}
```

5. Lakukan konfigurasi pada akun gmail yang digunakan dengan cara set pilihan Allow less secure apps: ON.



**Google Account**

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**Security**

Settings and recommendations to help you keep your account secure

**You have security recommendations**

Recommended actions found in the Security Checkup

[Protect your account](#)

**Recent security activity**

No security activity or alerts in the last 28 days

**Signing in to Google**

Password

Last changed Jul 3, 2021

**Google Account**

Less secure app access

Some apps and devices use less secure sign-in technology, which makes your account vulnerable. You can turn off access for these apps, which we recommend, or turn it on if you want to use them despite the risks. Google will automatically turn this setting OFF if it's not being used. [Learn more](#)

Allow less secure apps: ON

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